5. Suppose P, Q, and R are network service providers with respective CIDR address allocations C1.0.0.0/8, C2.0.0.0/8, and C3.0.0.0/8. Each provider's customers initially receive address allocations that are a subset of the provider's. P has the following customers:

PA, with allocation C1.A3.0.0/16

PB, with allocation C1.B0.0.0/12.

Q has the following customers:

QA, with allocation C2.0A.10.0/20

QB, with allocation C2.0B.0.0/16.

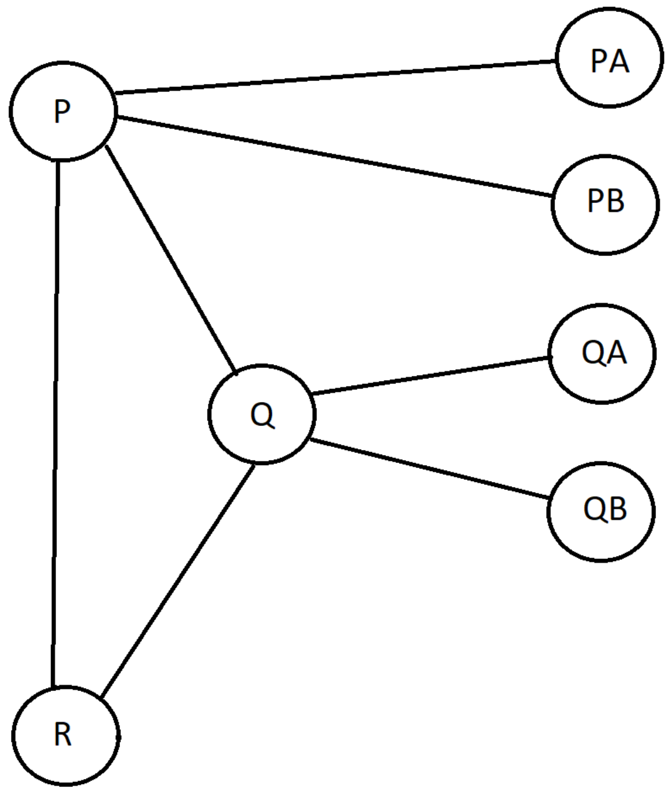
Assume there are no other providers or customers.

(a) Give routing tables for P, Q, and R assuming each provider connects to both of the others.

(b) Now assume P is connected to Q and Q is connected to R, but P and R are not directly connected. Give tables for P and R.

(c) Suppose customer PA acquires a direct link to Q, and QA acquires a direct link to P, in addition to existing links. Give tables for P and Q, ignoring R.

a. It's easier if you draw a diagram of the network:



P, Q, and R are network service providers.

CIDR Address Allocations:

P - C1.0.0.0/8

Q - C2.0.0.0/8

R - C3.0.0.0/8

P has the following customers:
PA, with allocation C1.A3.0.0/16
PB, with allocation C1.B0.0.0/12.
Q has the following customers:
QA, with allocation C2.0A.10.0/20
QB, with allocation C2.0B.0.0/16.

P's table:

Address	Next Hop
C2.0.0.0/8	Q
C3.0.0.0/8	R
C1.A3.0.0/16	PA
C1.B0.0.0/12	PB

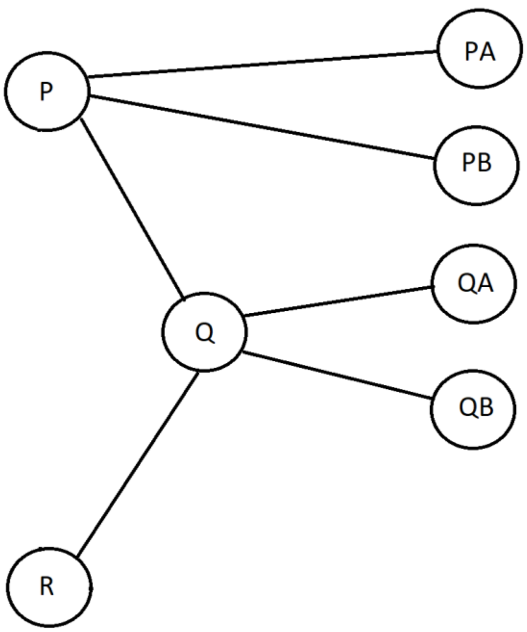
Q's table:

Address	Next Hop
C1.0.0.0/8	P
C3.0.0.0/8	R
C2.0A.10.0/20	QA
C2.0B.0.0/16	QB

R's table:

Address	Next Hop
C1.0.0.0/8	P
C2.0.0.0/8	Q

b. Diagram:



P is connected to Q
Q is connected to R
P and R are not directly connected
Give tables for P and R. Q's table doesn't change anyway.

P's table:

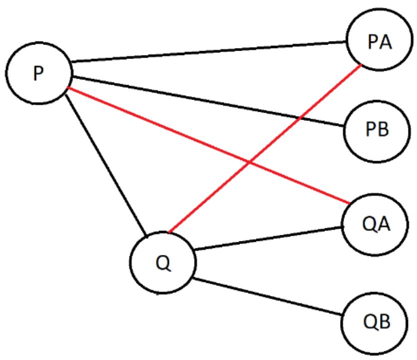
Address	Next Hop
C2.0.0.0/8	Q
C3.0.0.0/8	Q
C1.A3.0.0/16	PA
C1.B0.0.0/12	PB

R’s table:

Address	Next Hop
C1.0.0.0/8	Q
C2.0.0.0/8	Q

c. Suppose customer PA acquires a direct link to Q, and QA acquires a direct link to P, in addition to existing links. Give tables for P and Q, ignoring R.

Diagram, red links are the new direct links. We ignore R because that’s what the problem says to do:



PA directly connects to Q
QA directly connects to P

P has the following customers:

- PA, with allocation C1.A3.0.0/16
- PB, with allocation C1.B0.0.0/12.

Q has the following customers:

- QA, with allocation C2.0A.10.0/20
- QB, with allocation C2.0B.0.0/16.

P - C1.0.0.0/8

Q - C2.0.0.0/8

R - C3.0.0.0/8

P’s table:

Address	Next Hop
C2.0.0.0/8	Q
C2.0A.10.0/20	QA
C1.A3.0.0/16	PA
C1.B0.0.0/12	PB

Q’s table:

Address	Next Hop
C1.0.0.0/8	P
C1.A3.0.0/16	PA
C2.0A.10.0/20	QA
C2.0B.0.0/16	QB